TE 10 Mode

Aim: To simulate the dominant TE_{10} mode in a WR-90 rectangular waveguide using HFSS and determine the electric and magnetic field distributions, cutoff frequency, and propagation constant, comparing the simulated cutoff frequency with the theoretical value.

Software required:
Matlab

Program:

```
clc;
clear;
close all;

%% Waveguide Parameters
a = 22.86e-3;       % Width (m)
b = 10.16e-3;       % Height (m)
L = 100e-3;         % Length (m)
f = 10e9;           % Operating frequency (Hz)
c = 3e8;            % Speed of light (m/s)
mu0 = 4*pi*1e-7;    % Permeability of free space
eps0 = 1/(mu0*c^2); % Permittivity of free space
omega = 2*pi*f;

%% Theoretical Calculations
fc = c/(2*a);
beta = (2*pi*f/c)*sqrt(1-(fc/f)^2);
lambda_g = 2*pi/beta;

fprintf('TE10 Waveguide Simulation\n');
fprintf('Cutoff frequency = %.4f GHz\n',fc/1e9);
fprintf('Operating frequency = %.2f GHz\n',f/1e9);
fprintf('Propagation constant = %.4f
rad/m\n',beta);
fprintf('Guide wavelength = %.4f
mm\n',lambda_g*1000);

%% Spatial Grid
x = linspace(0,a,100);
y = linspace(0,b,50);
z = linspace(0,L,100);
[X,Y] = meshgrid(x,y);

%% TE10 Electric Field (Ez = 0, Ey is dominant)
Ey = sin(pi*X/a);
Ex = zeros(size(X));

figure;
imagesc(x*1000,y*1000,Ey);
set(gca,'YDir','normal');
colorbar;
xlabel('Width x (mm)');
ylabel('Height y (mm)');
title('TE_{10} Electric Field Distribution (E_y)');
axis equal tight;

%% Magnetic Field Components (normalized)
% For TE10: Hx and Hz components
[XX,ZZ] = meshgrid(x,z);
Hx = -(beta/(omega*mu0))*cos(pi*XX/a);
Hz = (pi/(a*omega*mu0))*sin(pi*XX/a);

figure;
subplot(1,2,1);
imagesc(x*1000,z*1000,Hx);
set(gca,'YDir','normal');
colorbar;
xlabel('Width x (mm)');
ylabel('Length z (mm)');
title('H_x Distribution');

subplot(1,2,2);
imagesc(x*1000,z*1000,Hz);
set(gca,'YDir','normal');
colorbar;
xlabel('Width x (mm)');
ylabel('Length z (mm)');
title('H_z Distribution');

%% 3D Electric Field Variation Along Length
[X3,Z3] = meshgrid(x,z);
E3 = sin(pi*X3/a).*cos(beta*Z3);

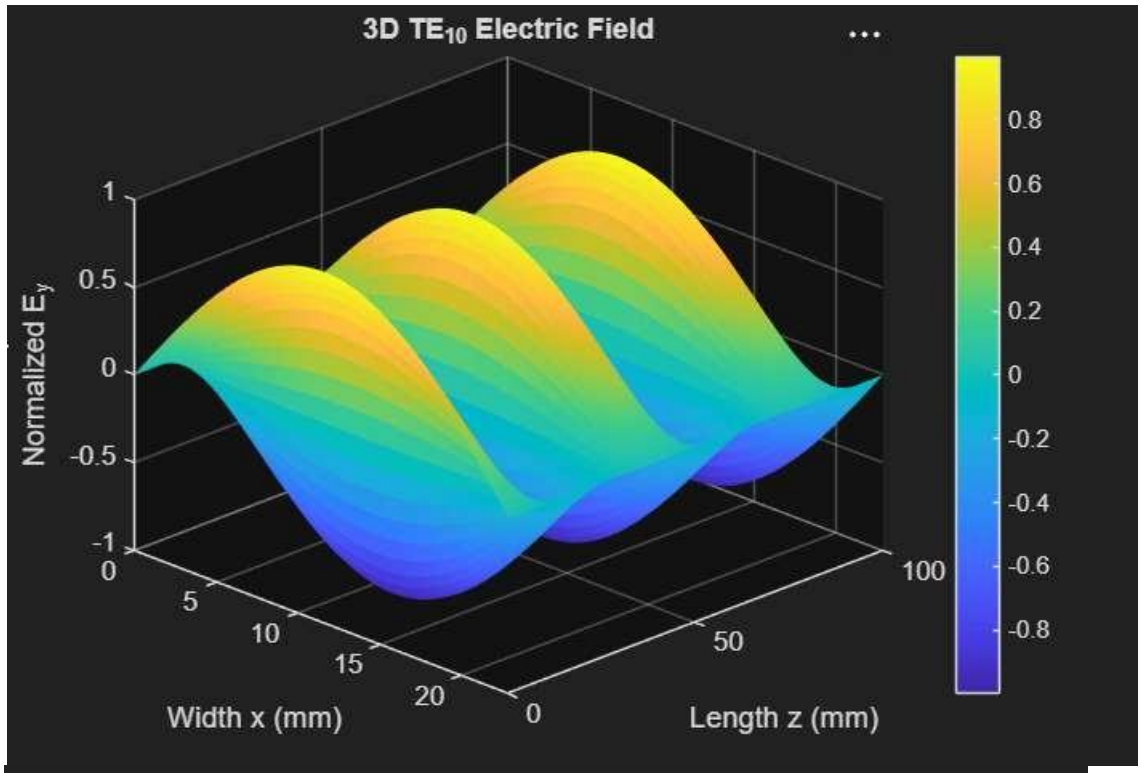
figure;
surf(X3*1000,Z3*1000,E3,'EdgeColor','none');
colorbar;
xlabel('Width x (mm)');
ylabel('Length z (mm)');
zlabel('Normalized E_y');
title('3D TE_{10} Electric Field');
view(45,30);

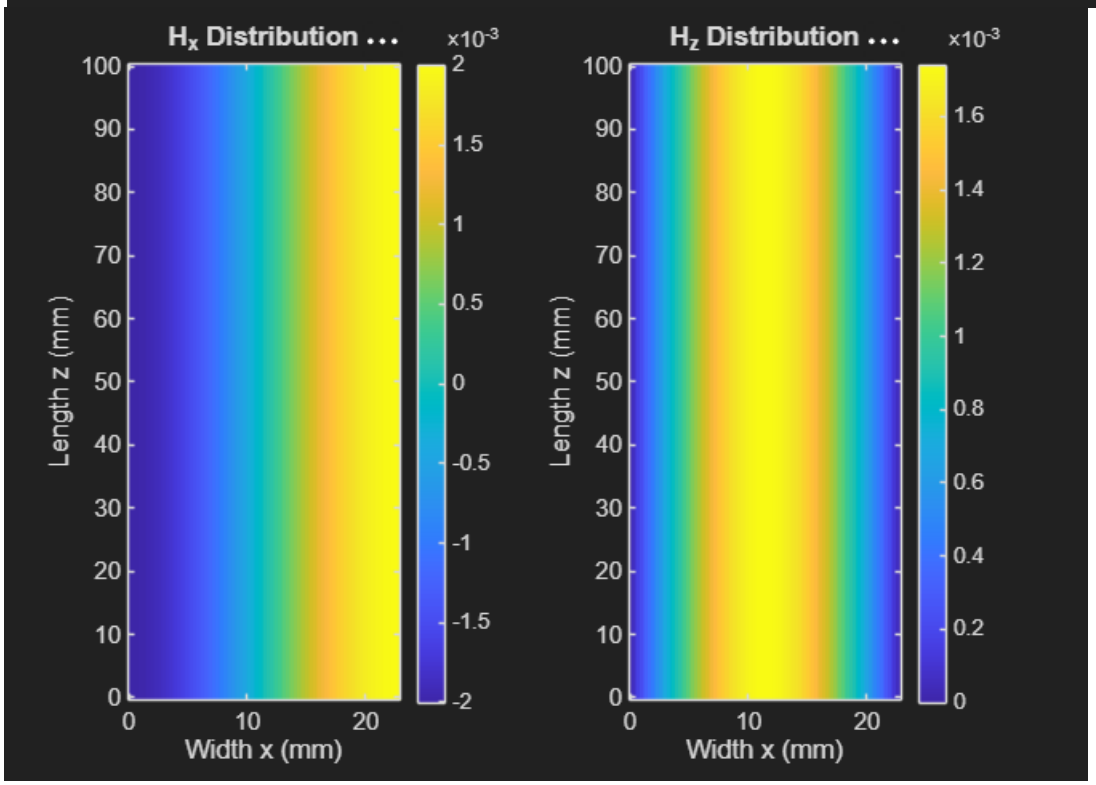
%% Simulated Cutoff Frequency Input
fc_sim = input('Enter simulated cutoff frequency in
GHz: ');
fc_sim = fc_sim*1e9;

deviation = abs(fc_sim-fc)/fc*100;

fprintf('Theoretical cutoff frequency = %.4f
GHz\n',fc/1e9);
fprintf('Simulated cutoff frequency = %.4f
GHz\n',fc_sim/1e9);
fprintf('Percentage deviation = %.4f
%%\n',deviation);
```


OUTPUT:


3D TE10 Electric Field
Normalized Ey
Width x (mm)
Length z (mm)


Hx Distribution
Hz Distribution
Length z (mm)
Width x (mm)
x10-3

```
TE10 Waveguide Simulation
Cutoff frequency = 6.5617 GHz
Operating frequency = 10.00 GHz
Propagation constant = 158.0462 rad/m
Guide wavelength = 39.7554 mm
Enter simulated cutoff frequency in GHz: 6.517
Theoretical cutoff frequency = 6.5617 GHz
Simulated cutoff frequency = 6.5170 GHz
Percentage deviation = 0.6809 %
```